COMPATIBILITY OF THE TRACE FOR ARTIN-SCHREIER LIKE EXTENSIONS WITH ALGEBRA MAPS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file resolve.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the reduction to the universal map over a polynomial ring in the coefficients with the compatibility of the trace maps, the verification that the relevant coefficients vanish for degrees below p minus one by recognizing a derivative, and the explicit computation of the top coefficient modulo p using the vanishing of the intermediate binomial coefficients.

Editorial note. The label lemma-trace-well-defined names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source resolve.tex –label lemma-trace-well-defined –start 399 –end 498 –local-id 28069 returned RESULT=OK (L28069 free, occupancy FREE, public interval clear). Candidate taken from sources/stacks-resolve/candidates-halfA.json.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Universal construction reducing to polynomial coefficients. Standard reduction, used explicitly in the proof, passing from an arbitrary relation in the ring R to the universal polynomial ring in the coefficients. Supplies the universal algebra map between the two quotients of polynomial rings and the commutative square reducing the statement to the universal case.
- Formal differentiation and the behaviour of the derivative on p-th powers. Algebra, the calculus of the formal derivative, in particular the vanishing of the derivative of a p-th power in characteristic p and the rules used in the two computations. Supplies the rewriting of the expressions as derivatives and the computation of the coefficients modulo p.

Lemma 0.1. *Let $\varphi : R[x]/(x^p - a) \to R[y]/(y^p - b)$ be an R-algebra homomorphism. Then $Tr_x = Tr_y \circ \varphi$.*

Proof. Say $\varphi(x) = \lambda_0 + \lambda_1 y + \ldots + \lambda_{p-1} y^{p-1}$ with $\lambda_i \in R$. The condition that mapping x to $\lambda_0 + \lambda_1 y + \ldots + \lambda_{p-1} y^{p-1}$ induces an R-algebra homomorphism $R[x]/(x^p - a) \to R[y]/(y^p - b)$ is equivalent to the condition that

$$a = \lambda_0^p + \lambda_1^p b + \ldots + \lambda_{p-1}^p b^{p-1}$$

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in the ring R. Consider the polynomial ring

$$R_{univ} = \mathbf{F}_p[b, \lambda_0, \ldots, \lambda_{p-1}]$$

with the element $a = \lambda_0^p + \lambda_1^p b + \ldots + \lambda_{p-1}^p b^{p-1}$ Consider the universal algebra map $\varphi_{univ} : R_{univ}[x]/(x^p - a) \to R_{univ}[y]/(y^p - b)$ given by mapping x to $\lambda_0 + \lambda_1 y + \ldots + \lambda_{p-1} y^{p-1}$. We obtain a canonical map

$$R_{univ} \longrightarrow R$$

sending b, λ_i to b, λ_i. By construction we get a commutative diagram

$$\begin{array}{ccc} R_{univ}[x]/(x^p - a) & \longrightarrow & R[x]/(x^p - a) \\ \downarrow{\scriptstyle \varphi_{univ}} & & \downarrow{\scriptstyle \varphi} \\ R_{univ}[y]/(y^p - b) & \longrightarrow & R[y]/(y^p - b) \end{array}$$

and the horizontal arrows are compatible with the trace maps. Hence it suffices to prove the lemma for the map φ_{univ}. Thus we may assume $R = \mathbf{F}_p[b, \lambda_0, \ldots, \lambda_{p-1}]$ is a polynomial ring. We will check the lemma holds in this case by evaluating $\mathrm{Tr}_y(\varphi(x)^i \mathrm{d}\varphi(x))$ for $i = 0, \ldots, p-1$.

The case $0 \leq i \leq p-2$. Expand

$$(\lambda_0 + \lambda_1 y + \ldots + \lambda_{p-1} y^{p-1})^i (\lambda_1 + 2\lambda_2 y + \ldots + (p-1)\lambda_{p-1} y^{p-2})$$

in the ring $R[y]/(y^p - b)$. We have to show that the coefficient of y^{p-1} is zero. For this it suffices to show that the expression above as a polynomial in y has vanishing coefficients in front of the powers y^{pk-1}. Then we write our polynomial as

$$\frac{\mathrm{d}}{(i+1)\mathrm{d}y}(\lambda_0 + \lambda_1 y + \ldots + \lambda_{p-1} y^{p-1})^{i+1}$$

and indeed the coefficients of y^{kp-1} are all zero.

The case $i = p-1$. Expand

$$(\lambda_0 + \lambda_1 y + \ldots + \lambda_{p-1} y^{p-1})^{p-1} (\lambda_1 + 2\lambda_2 y + \ldots + (p-1)\lambda_{p-1} y^{p-2})$$

in the ring $R[y]/(y^p - b)$. To finish the proof we have to show that the coefficient of y^{p-1} times $\mathrm{d}b$ is $\mathrm{d}a$. Here we use that R is S/pS where $S = \mathbf{Z}[b, \lambda_0, \ldots, \lambda_{p-1}]$. Then the above, as a polynomial in y, is equal to

$$\frac{\mathrm{d}}{p\mathrm{d}y}(\lambda_0 + \lambda_1 y + \ldots + \lambda_{p-1} y^{p-1})^p$$

Since $\frac{\mathrm{d}}{\mathrm{d}y}(y^{pk}) = pky^{pk-1}$ it suffices to understand the coefficients of y^{pk} in the polynomial $(\lambda_0 + \lambda_1 y + \ldots + \lambda_{p-1} y^{p-1})^p$ modulo p. The sum of these terms gives

$$\lambda_0^p + \lambda_1^p y^p + \ldots + \lambda_{p-1}^p y^{p(p-1)} \bmod p$$

Whence we see that we obtain after applying the operator $\frac{\mathrm{d}}{p\mathrm{d}y}$ and after reducing modulo $y^p - b$ the value

$$\lambda_1^p + 2\lambda_2^p b + \ldots + (p-1)\lambda_{p-1} b^{p-2}$$

for the coefficient of y^{p-1} we wanted to compute. Now because $a = \lambda_0^p + \lambda_1^p b + \ldots + \lambda_{p-1}^p b^{p-1}$ in R we obtain that

$$\mathrm{d}a = (\lambda_1^p + 2\lambda_2^p b + \ldots + (p-1)\lambda_{p-1}^p b^{p-2})\mathrm{d}b$$

in R. This proves that the coefficient of y^{p-1} is as desired. $\square$